



AMAN RAJ

Enrollment No.: BT22CSE115

B.Tech – Computer Science and Engineering
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GitHub Profile

EDUCATION

Degree/Course	Institute / Board	CGPA/%	Year
B.Tech CSE	Visvesvaraya National Institute of Technology, Nagpur	6.01	2026
CLASS XII (CBSE)	J P Central School, Samastipur, Bihar	67.8%	2022
CLASS X (CBSE)	Green Valley International School, Pusa, Samastipur, Bihar	78.2%	2020

PROJECTS

Muzic - Music Streaming Website

Oct 2025

React, MongoDB, Express, Node.js

- Developed a full-stack music streaming web application using the MERN stack with secure user authentication and a responsive user interface.
- Implemented music playback, song search, and song management features using RESTful APIs and MongoDB for efficient data storage and retrieval.
- Deployed the frontend on Vercel and maintained the project with GitHub, ensuring scalable deployment, version control, and clean project organization.

AI News Summarizer

April 2025

GPT4All, Streamlit

- Developed an AI-powered web app using Streamlit to fetch, summarize, and evaluate news articles using GPT4All and DeepSeek LLMs.
- Integrated multi-agent architecture with a local LLM summarizer and OpenRouter-based quality checker to ensure accuracy in AI-generated content.
- Secured API keys via .env and structured the project for open-source deployment with GitHub, including clean documentation and dependency management.

Topology-Aware Reinforcement Learning for Adaptive Cooperative Caching in CCN

April 2026

Python — Deep Q-Network (DQN) — Reinforcement Learning — Network Simulation

- Developed a topology-aware reinforcement learning framework using Deep Q-Network (DQN) and centrality-based (CMBA) features to optimize cooperative caching in Content-Centric Networks (CCN).
- Designed and evaluated four topology migration scenarios to analyze policy adaptation, router selection, reward convergence, and latency performance under changing network conditions.
- Improved caching efficiency by implementing fine-tuning of pretrained models, achieving faster adaptation, lower latency, and more stable performance compared to training from scratch.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS, MySQL

Frameworks/Libraries: React.js, Node.js, Express.js, Streamlit

Databases: MongoDB, MySQL

Developer Tools: Git, GitHub, VS Code, Docker, Postman

Core Concepts: Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks, Machine Learning, Reinforcement Learning, Large Language Models (LLMs), REST APIs